

Character Stylization in Disney's "Wish"

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Figure 1: Final character stylization results (left-right): Stylized fur on Valentino. Tailoring on Magnifico evokes a graphic read of the costume. Linework on Asha's and Sabino's faces suggests the characters' age. Feather stylization on an owl.

ABSTRACT

"Wish" pays homage to a century of Disney Animation's legacy and the look of its characters is inspired by the watercolor storybook style of some of our earlier Disney classics, including "Snow White and the Seven Dwarfs", "Pinocchio", and "Sleeping Beauty". Careful attention to detail in appearance, graphic shading, and streamlined geometric shapes of models, clothing and grooms, provide the foundation of the unique stylized look and art-directed character performance. A final procedural compositing treatment is applied in lighting to produce the more painterly watercolor style combined with hand-drawn influenced linework. Modifications were required in all of the asset departments' workflows to support this overall stylization.

KEYWORDS

animation, simulation, 2D linework, nonphotorealistic rendering

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1 INTRODUCTION

For "Wish", the desire was to retain Disney's classic sense of character appeal, influenced by a legacy of hand-drawn animation, while incorporating stylized elements of watercolor and linework inspired

by our earlier films. In addition, the unique stylization needed to authentically represent our ethnically diverse cast of characters. For our modern films, Disney typically produces a 3D aesthetic, and our current tools and workflows have been designed to support this type of look. To achieve all this, it was necessary to alter the way our assets were built to capture the base stylized art-direction. The final result, including painterly qualities and hand-drawn influenced linework, was then achieved via *stylization modifiers*, a custom compositing process applied in lighting driven by arbitrary output variables (AOVs) generated in the render. The novel approach developed on "Wish" was a concentrated collaboration between all of the film production departments with a large amount of testing, feedback, and adjustments to our overall pipeline and tool sets.

2 APPROACH

Starting early in the pre-production process, the creative leadership, artistic, and technical teams iterated for months to co-create the unique stylized look for our film. Through this effort, several advancements were made to the asset and lighting departments' existing workflows that helped us achieve the overall stylization for our characters. The Look and Lighting teams also developed a render time diagnostic tool that made it possible to continuously review the character at various distances to make sure the stylization of characters held up across expected ranges.

2.1 Character Asset Workflows

The procedural stylization modifiers in the compositing phase produce 2D linework where hard edges or creases occur in the geometry. Therefore in both modeling and rigging, close attention was given to plane changes and topology. Models were analyzed both in static poses and animated performance, then careful adjustments were made to shape, curvature, and weighting of characters to proactively influence the final stylized linework (see Figure 1).

The base look of the character had to be significantly altered to stay aligned with the final stylized appearance. Textures were

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strategically designed with levels of detail for representing components like dirt and wear. Complex structures like weaves and seams were carefully limited and caricatured.

For faces, varying amounts of shaping contours were painted into surface color in place of specular highlights. It is important that the skin colors stay true to the diverse ethnicities, while still feeling like the characters belonged together in the stylized world. Overall, surface color was unified to provide an illustrated look, giving space for the stylization tools to later introduce dynamic hue variation in a manner more in line with watercolor. Materials are designed to downplay sheen and shine which inherently emphasize the dimensionality of an object. Specular highlights and core scatter were reduced for fur and hair materials, in order to achieve a flatter, two-dimensional effect.

Grooms were sculpted into a solid silhouette with graphic clump shapes, with reduced hair density and increased hair width. Hair texture details like noise, strays, length variation, and taper were adjusted to make shapes more defined and to ultimately produce the desired linework from the stylization modifiers (see Figure 2). Peach fuzz on skin was avoided to limit surface breakup. Our typical feather grooming workflow was adjusted to use a combination of textures, geometry, and fur, to more subtly suggest the material instead of grooming individual feathers (e.g. the owl in Figure 1).

For cloth, tailoring workflows were evolved to produce garments with the desired amount of shape detail in the drapery and folds in order to evoke a graphic read of the costume, which supported the needs of stylization. Simulation rigs were designed to ensure that this art-directed drapery stayed on model, through all levels of characters' physical activity and motion (see Figure 1). Strategic choices were made to limit the amount of folds and wrinkles produced by the simulation, in order for the final linework to intentionally appear in desired areas only.

Hair simulation rigs were crafted so that the groom breaks into specifically art-directed shapes that support the stylized groupings of hair in motion.

Our designers were inspired by the visual style of crowd groupings from the Disney classics. To achieve this effect, when crowds recede into the distance, shapes are simplified, detail is reduced, and lines are offset from bodies. Crowd groups are clumped into sections of color, with the intention of becoming a part of the environmental



Figure 3: Base render (left). Final result including modifiers (right).

architecture, like a gestural swatch of painterly hue. Advancements to our crowd look workflows allowed the team to meticulously choose at a shot or sequence level, the casting of crowd characters to support the specific color schemes and stylization.

2.2 Lighting Stylization Workflows

As a final procedural stylization pass, a new compositing workflow was developed using the Nuke toolset which was packaged into a set of stylization modifiers. These modifiers are applied on top of the base stylized look realized for the characters, adding linework and painterly qualities based on the carefully underlying asset structures (see Figures 1, 3). Depth lines are generated from edge detection of geometry and include major silhouettes, whereas normal lines are generated from edge detection of surface normals, emphasizing internal detail. Additionally, custom lines are defined through an RGB map, such as a groom's interpolation or clumping region map. Line color is further modified with a map or expression. Finally, watercolor and paper textures are applied globally to all assets. These settings are adjusted as needed by the character look development team, with mattes and custom dialed with individual controls.

Look collaborated closely with Lighting to iterate on the modifiers to come up with a system for handling unique cases in shots. Sometimes this involved using mattes to control line placement or to blend details into watercolor texture.

Lines are an integral part of the overall stylization. Our Production Design and Animation teams envisioned lines as a means of accentuating important aspects of a character's appearance, expressions, and performance. Calisthenic tests were generated to evaluate characters as they performed. Adjustments were accordingly made to the stylization modifiers by restricting masks and changing line contrast and color. Linework has an immense effect on the perceived age of the character, most notably for lip, hand, and face lines (e.g. see Figure 1). There are subtleties to linework that are more straightforward to control via hand-drawn methods. To capture those subtleties in CG, the Look Development team appended new modifiers as needed for additional lines, such as on a character's knuckles, fingers, wrinkles etc. Lighting could manually enable these for closeup shots, as this detail was desired in certain situations.

3 CONCLUSIONS AND FUTURE WORK

For Disney's "Wish", as a way to honor the studio's past, the character team found ways to evolve asset workflows and closely collaborated to create new tools and techniques to support a watercolor-inspired illustrative look. The resulting workflows are already in use in other areas of our pipeline. The "Wish" experience serves as a foundation to inspire new looks and techniques on future projects.

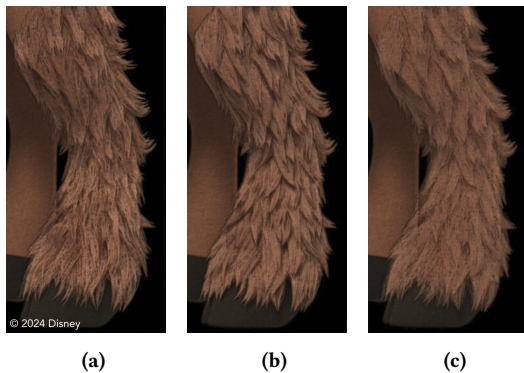


Figure 2: Early fur grooming and shading tests experimenting with (a) standard look, (b) increased density/rougher reflectance, and (c) finer hair width/increased noise.